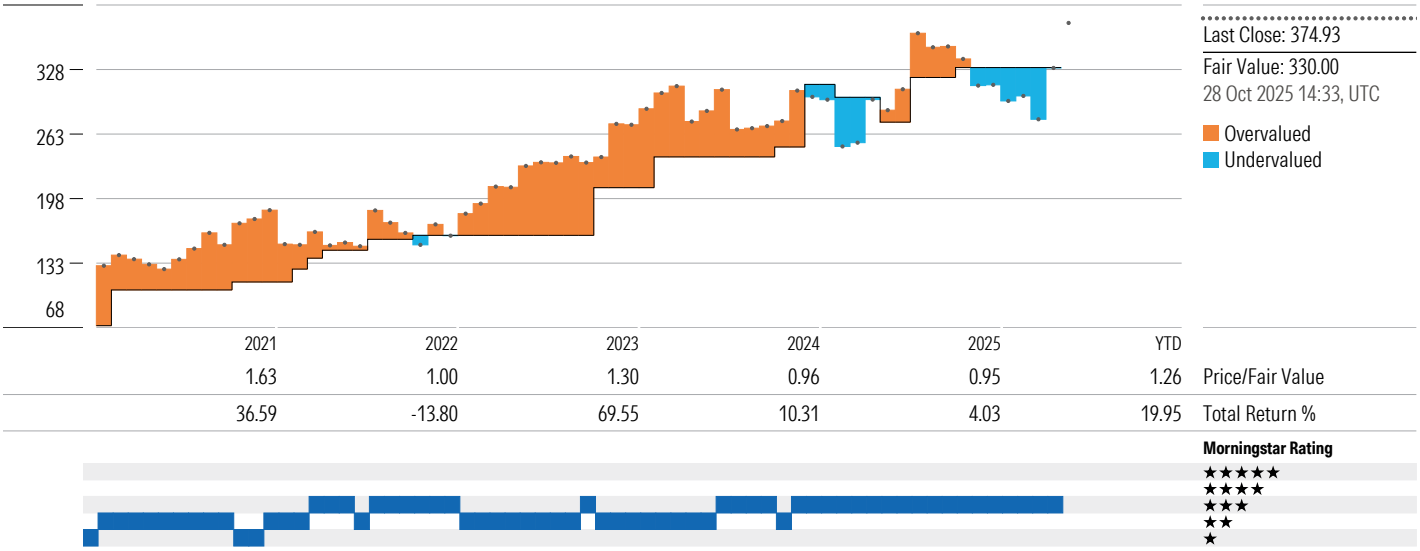
Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Price vs. Fair Value



	2021	2022	2023	2024	2025	YTD
Price/Fair Value	1.63	1.00	1.30	0.96	0.95	1.26
Total Return %	36.59	-13.80	69.55	10.31	4.03	19.95

Morningstar Rating

Total Return % as of 31 May 2026. Last Close as of 31 May 2026. Fair Value as of 28 Oct 2025 14:33, UTC.

Contents

Analyst Note (28 Apr 2026)
Business Description
Business Strategy & Outlook (28 Apr 2026)
Bulls Say / Bears Say (28 Apr 2026)
Economic Moat (28 Apr 2026)
Fair Value and Profit Drivers (28 Apr 2026)
Risk and Uncertainty (28 Apr 2026)
Capital Allocation (28 Apr 2026)
Analyst Notes Archive
Financials
ESG Risk
Appendix
Research Methodology for Valuing Companies

Important Disclosure

The conduct of Morningstar's analysts is governed by Code of Ethics/Code of Conduct Policy, Personal Security Trading Policy (or an equivalent of), and Investment Research Policy. For information regarding conflicts of interest, please visit: http://global.morningstar.com/equitydisclosures.

The primary analyst covering this company does not own its stock.

'The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

Cadence Earnings: Record Backlog and Agentic AI Momentum Drive Outlook Raise

Analyst Note Eric Compton, CFA, Director, 28 Apr 2026

Cadence reported a powerhouse first quarter to start 2026, with revenue growing 19% year over year and a record backlog of $8 billion. The firm raised its full-year revenue growth outlook to 17%.

Why it matters: We have consistently reiterated that we expect EDA firms, like Cadence, to be beneficiaries of the artificial intelligence era, and this quarter provided further evidence. Cadence launched its AgentStack framework, moving beyond simple AI-enabled features to autonomous "Super Agents."

- Agents can automate previously manual tasks, such as RTL coding, while improving engineers' productivity. While management has not provided exact pricing details, they did reference a "subscription plus consumption" framework, and we see AI driving heightened consumption.
- Cadence is also increasingly positioning itself for the physical AI era, with its recent acquisition of Hexagon's design and engineering business. This is yet another material, multiyear revenue driver we see for EDA firms. Further related to AI demand was another record quarter of hardware sales.

The bottom line: We maintain our fair value estimate of $330 for wide-moat Cadence and view the stock as fairly valued. Despite software stocks being down materially year to date, Cadence is up roughly 9%.

- The quarter was a bit messy given the recent Hexagon acquisition; however, when backing out the

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Sector Technology

Industry Software - Application

Business Description

Cadence Design Systems is a provider of electronic design automation software, intellectual property, and system design and analysis products. EDA software automates and aids in the chip design process, enhancing design accuracy, productivity, and complexity in a full-flow end-to-end solution. Cadence offers a portfolio of design IP, as well as system design and analysis products, which enables system-level analysis and verification solutions.

effects of this, management actually raised its full-year EPS guidance by $0.08, or roughly 10%. Hexagon is expected to be accretive by 2027.

- China-related revenue continues to be stronger than we would have anticipated a year ago, remaining at 13% of total revenue. Without material deterioration here, we would not be surprised to see an additional guidance raise before the year is over, given the strength in the first quarter.

Big picture: Given the momentum with AI demand, along with the upcoming demand for the physical AI buildout, we think midteens revenue growth is quite likely for the foreseeable future.

Business Strategy & Outlook Eric Compton, CFA, Director, 28 Apr 2026

Cadence Design Systems is a leading provider of software tools used by engineers to design semiconductor chips. This software is referred to as electronic design automation software. EDA solutions are indispensable to major semiconductor design firms, and increasingly a growing number of “electronic systems” companies that design a product that uses semiconductors, such as modern automobiles. The semiconductor industry does not function without companies like Cadence in the background.

Cadence is one of three major EDA players, along with Synopsys and Siemens EDA. Cadence is historically stronger in the analog space, whereas Synopsys has historically been more dominant in the digital space. But Cadence in particular has increased its share within digital workflows over the years.

Cadence has three business segments: core EDA, intellectual property, and system design and analysis. We believe the outlook is strong for all three. Demand for EDA tools is set to grow significantly over the next decade as artificial intelligence demand increases, design starts, and chip complexity in general increases (such as multi-die and chiplet architectures), and nonsemiconductor firms that use semiconductors in their products increasingly use EDA tools. We also see Cadence monetizing the productivity and performance results that its AI tools provide.

The intellectual property segment sells predesigned chip components. We expect above-average growth here, as well as the industry needs prebuilt IP blocks to speed up the design process and focus efforts on areas where design can be differentiated.

We like how Cadence is positioned in system design and analysis, as it has developed many of its key simulation and systems design tools in-house. We think this is the future of the industry, as EDA tools and high-end design needs proliferate beyond the core semiconductor industry.

From a strategic perspective, we think Cadence will see above-average growth from its systems strategy. Its focus on innovation, such as its improved AI offerings, should serve it well. Cadence should remain a dominant force in the industry for years to come, while margins gradually increase with growing revenue.

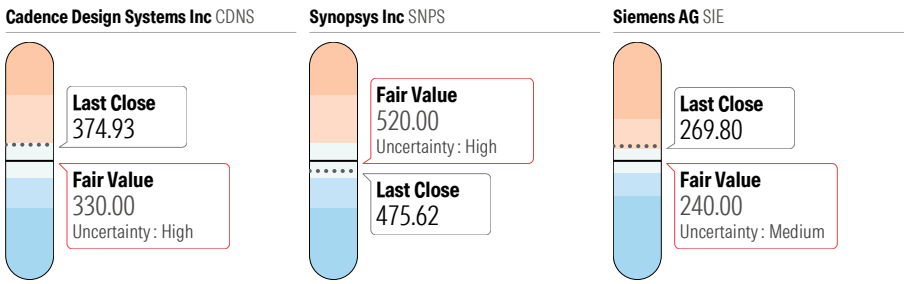
© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Competitors



	Cadence Design Systems Inc CDNS	Synopsys Inc SNPS	Siemens AG SIE
Economic Moat	Wide	Wide	Wide
Currency	USD	USD	EUR
Fair Value	330.00 1 Jun 2026 17:17, UTC	520.00 1 Jun 2026 17:23, UTC	240.00 29 May 2026 23:25, UTC
1-Star Price	511.50	806.00	324.00
5-Star Price	198.00	312.00	168.00
Assessment	Overvalued 1 Jun 2026	Fairly Valued 1 Jun 2026	Overvalued 1 Jun 2026
Morningstar Rating	★★1 Jun 2026 05:00, UTC	★★★1 Jun 2026 05:00, UTC	★★29 May 2026 05:00, UTC
Analyst	Eric Compton, Director	Eric Compton, Director	Matthew Donen, Director
Capital Allocation	Exemplary	Exemplary	Exemplary
Price/Fair Value	1.26	0.95	1.12
Price/Sales	18.54	10.05	2.67
Price/Book	15.76	2.99	3.20
Price/Earning	87.40	110.10	27.90
Dividend Yield	0.00%	0.00%	1.98%
Market Cap	114.23 Bil	94.26 Bil	207.96 Bil
52-Week Range	262.75—414.84	376.18—651.73	198.00—279.60
Investment Style	Large Growth	Large Blend	Large Growth

Bulls Say Eric Compton, CFA, Director, 28 Apr 2026

- With limited competition and products that are essential in support of the semiconductor industry, which generates more than $500 billion in annual revenue, Cadence's moat is wide and durable.
- Secular tailwinds such as physical AI and AI tools will drive new chip design starts, raise R&D intensity, and open up new markets for semiconductor-focused EDA tools, leading to excess revenue growth for 10 or more years.
- Cadence' margins should only improve, supported by improving IP margins, greater AI-driven wallet share, and internal AL/ML optimizations that enhance operational efficiency.

Bears Say Eric Compton, CFA, Director, 28 Apr 2026

- The valuation is not cheap, and the market is already giving Cadence full credit for the upcoming growth cycle. Any hiccups along the way will lead to declines in valuation.

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

- Geopolitical tensions with China and China's focus on supporting domestic companies could affect Cadence's ability to do business in the region and lead to slower revenue growth.
- Cadence has less share with the overall IP market, and currently the analog space is not growing as fast as the digital semiconductor industry given the AI demand boom, so Cadence could underperform peers.

Economic Moat Eric Compton, CFA, Director, 28 Apr 2026

We believe Cadence warrants a wide moat rating due to switching costs and intangible assets associated with its vast product suite. Cadence provides leading-edge electronic design automation tools and intellectual property critical to the semiconductor chip design process. We believe that Cadence enjoys high customer switching costs and intangible assets in its sticky EDA segment, as well as intangible assets associated with its IP offerings. In our opinion, this customer stickiness and profitable IP portfolio will drive excess returns on capital for Cadence over the next 10-20 years. Fundamentally, we believe that as long as the semiconductor industry exists and is innovating, Cadence will be a necessary part of that innovation, earning excess returns on invested capital.

Electronic design automation software is what engineers use to design and build semiconductor chips. It is indispensable, functioning as the backbone for developing advanced microchips. Without EDA software, there would be no modern semiconductor industry. If a company is designing semiconductor chips, or its overall systems are heavily reliant on semiconductors (think modern automobiles), it is using EDA software.

The EDA industry is dominated by three key players that control over 90% of the market: Synopsys, Cadence, and Siemens EDA (formerly Mentor Graphics). Among these three, Synopsys and Cadence are the most dominant market leaders, each controlling close to 40% of the market. This dominance is reinforced by near 100% retention rates and an 80%-85% recurring revenue model, reflecting the critical role that EDA tools play in chip development and the loyalty the top EDA firms have from their clients. These are essentially the strongest retention rates a software company can have, implying customer lifetimes well over 20 years. The primary reason a client stops using the services of Cadence is that it goes out of business. We believe the top EDA firms' market dominance (essentially a stable oligopoly), evidence of extremely high switching costs (retention rates near 100%), and position as indispensable tools for processes where firms cannot afford failure (a bad chip design could put a younger firm out of business, or at the very least have a severely negative impact on a more mature firm) all support a wide moat rating based on switching costs and intangible assets.

Switching costs represent the primary moat source for Cadence. Chip design is an intricate process involving thousands of hours dedicated to fine-tuning workflows, integrating tools, and aligning processes. Cadence's tools are tightly integrated with its clients' workflows, and transitioning to an alternative could lead to significant implementation risks, such as a drop in productivity and complex

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

data migration challenges. In a rapidly moving technological industry, we believe that the costs of switching vendors far outweigh the small technological benefit the client would get in switching vendors. In addition, engineers invest considerable time in becoming proficient with specific tools, and changing vendors would mean more training, additional ramp-up time, disruption to ongoing projects, and delayed time to market.

Further, firms cannot afford mistakes in the design process, as it puts future revenue and the future of the business at risk. Minor design flaws can result in significant disruptions and exacerbated losses down the line, making it necessary for these tools to be reliable and trustworthy. Clients would find it impractical to change providers without jeopardizing project deadlines and the quality of the final product.

In support of intangible assets, we point to the overall reputation of top EDA firms, like Cadence. When a company is considering which EDA tools to use, there are only three reliable choices. We find it highly unlikely that a company would entrust its design processes to an unknown, unproven upstart. It is simply not worth the risk. This hesitancy toward new vendors, along with gradual industry consolidation over time, has led to minimal investment in new alternatives. This is in stark contrast to other parts of the semiconductor industry, such as chip design (the clients of EDA firms), where new capital and companies jumped at the opportunity presented by new waves of demand, such as designing AI-optimized chips.

The barriers to entry in the EDA business are substantial, involving massive upfront capital, deep technical knowledge, and cumulative industry expertise that cannot be easily replicated. Synopsys' and Cadence's continuous acquisitions to build their software portfolio have led to market consolidation and a robust pace of innovation, which has put them leagues ahead of smaller competitors. We observe that if either lacks a feature or process, they have both the resources and technical know-how to catch up and maintain their competitive advantage. A major advantage that Cadence has over smaller competitors is its ability to offer end-to-end design flow—a full suite of tools needed to design a chip from start to finish—thus reducing the appeal of small competitors that have fragmented solutions.

Another industry feature that gives us confidence in our wide moat rating is the fact that EDA is not cyclical. During uncertain times and economic downcycles in the semiconductor market, firms can cut budgets, but one area where firms are loath to cut spending is their core R&D budget. Innovation is essential to maintaining a competitive edge in the semiconductor industry. If a company starts cutting its EDA budget significantly, it means it is essentially firing its engineers, which also likely means it is facing serious financial difficulty, likely even on the verge of going out of business.

Over the long term, we expect demand for EDA to only increase. As chips get smaller and more complex, and therefore more expensive to design, customers are willing to pay a premium for the latest tools that

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Cadence has to offer. Increasing complexity and increasing difficulties in chasing improved performance also increase the value of the solutions EDA firms offer. To meet this demand, Cadence has some of the highest R&D spending in the software space to ensure it remains at the cutting edge.

Another factor increasing our confidence in the long-term demand outlook is the increasing diversity of Cadence's client base, extending beyond semiconductor-focused companies to include what are called system-level customers. Fifteen years ago, nearly all of Cadence's clients were semiconductor companies; today, close to 45% are companies that produce the end products rather than the chips alone. For example, an automotive firm that has to design a car that uses many different semiconductor components will design the car with the overall system, including the semiconductors, in mind. They can use EDA software for this system-level design process. With products only becoming more complex and more electronic over time, we expect this demand trend will remain in place for decades to come. With the EDA industry consolidated around a few players, Cadence's brand and unique technological advantage position it to capture pricing uplifts as well as volume increase from a broadening customer base, along with industry-leading margins, all of which ultimately flow into better returns on capital.

Cadence's other primary business is its IP segment, which is becoming an increasingly important part of the company. As of fiscal 2025, IP growth had outpaced Cadence's core EDA growth, representing around 14% of the company's revenue.

Cadence's IP solutions can be compared with Lego building blocks. When designing a chip, it is helpful for engineers to use prebuilt blocks for parts of the chip that are common and undifferentiated, allowing designers to focus on customizing the unique, differentiated aspects of their creation. For example, part of the chip may have to deal with interacting with a USB connection. Building out this part of the chip each time a new chip is designed is not an efficient use of engineering resources. Companies like Cadence will prebuild this part of the chip and then sell that part of the design as IP. This allows clients to purchase these modular, proven, undifferentiated parts of a chip's design for a fraction of what it would cost them to develop them in-house with their own engineering team. Cadence can sell this IP at an attractive price because it is able to depreciate its cost of development over a large base of clients. Finally, speed to market is also important, and having an established IP business at the center of the design ecosystem allows Cadence to compete for the fastest time to market with verified IP blocks.

We believe this business benefits from an intangible asset moat source, driven by the breadth of its IP library and Cadence's unique position at the center of the semiconductor design workflow.

Regarding breadth, Cadence has steadily expanded its IP library (through acquisitions and internal development), building a robust repository of standardized and custom solutions that differentiate it from smaller competitors. The extensive library, combined with continuous innovation in sectors like

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

automotive and AI-focused applications keep Cadence at the forefront of client demands and the pace of technological developments. Nearly half of Cadence's workforce is dedicated to building and maintaining this library, a testament to the company's commitment to quality and breadth. This breadth makes Cadence a natural first place to look for IP solutions as a designer.

The second key ingredient to Cadence's moat in IP is its unique position at the center of semiconductor design. In theory, anyone could create many of these IP blocks. However, it is fundamentally more efficient at the industry level to have one or two firms do it, and then sell those standardized design blocks to the rest of the industry, as opposed to having each design firm rebuild what are functionally the same design blocks for each and every project. This allows a firm like Cadence, which is naturally already at the center of the design process for the industry with its software, to have a natural distribution advantage. This further allows Cadence to depreciate the cost of development for each IP block across a wider base of customers. We hesitate to award the firm a cost advantage for this because, in theory, it does not cost Cadence any less to develop these IP blocks, but the ability to sell them across a wider base of customers does allow Cadence to sell them at a price which would be unpractical for a firm with too few customers, and at a price which is cheaper than what it would cost any of its customers to develop these IP blocks in house.

Cadence grew its market share in the IP business to 6% in 2024, growing faster than the overall IP industry. Arm remains the dominant player overall, with 44% share, while Synopsys is number two with 24% share. The consistent quality of Cadence's IP fosters long-term partnerships and deepens client loyalty. Furthermore, client-specific customized solutions developed with Cadence make switching to other providers challenging, as it risks compromising design integrity and reliability.

Lastly, the type of IP portfolio matters. Arm has the most dominant market share, but it is focused only on processor IP. Meanwhile, Synopsys and Cadence concentrate on non-processor IP, which is growing fast, and we forecast it will soon account for more than half of the IP market. Given a fast-growing IP margin profile, we believe Cadence's IP business will be a significant contributor to strong and improving ROICs and future growth.

Fair Value and Profit Drivers Eric Compton, CFA, Director, 28 Apr 2026

Our fair value estimate of $330 per share implies a fiscal 2026 adjusted price/earnings multiple of 42 times and an enterprise value/sales multiple of 14 times.

Our fair value estimate assumes Cadence will be able to increase revenue at a 12% compound annual growth rate over the next decade, above the historical mid- to high-single-digit rate that the overall semiconductor industry has typically seen. This is based on increases in design starts from AI-driven designs, rising chip complexity leading to greater use of EDA tools, increased demand for IP blocks, a slowdown in Moore's law resulting in heightened demand for performance improvements through

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

design changes, and an increase in systems companies utilizing Cadence's offerings. We also expect Cadence will monetize the additional performance and efficiency value its AI tools bring, providing room for additional pricing power. Finally, we think the physical AI buildout will also be a key, long-term driver of demand.

Within our overall revenue growth projections, we assume that the core EDA business grows at a 10-year CAGR of 11%, IP grows at 13%, and systems design grows at 14%.

We also expect operating margin to continually increase for Cadence, from 31% in 2025 to 38% by 2035. We see no reason to expect margin expansion to stop by 2035, either. This will be driven by strong growth in the IP business (where we expect margins to improve as the business scales), AI-driven efficiencies, pricing opportunities through selling higher-value-added products (value-add increases driven by increased demand along with product innovation, like AI enhancements), and more typical scale based efficiencies (software companies do not need to double the salesforce or G&A to double sales, for example).

A stable business model and low cyclicality in revenue because of resilient demand during economic downturns give us confidence in assigning a 7.6% cost of equity.

Risk and Uncertainty Eric Compton, CFA, Director, 28 Apr 2026

We assign Cadence a High Morningstar Uncertainty Rating. This is primarily based on the uncertainty of future revenue growth. While the EDA firms have historically grown closer to the overall semiconductor market, they are entering a period where we expect above-industry growth for the next two decades. The degree of uncertainty around this revenue upside and its implications for our fair value estimate leads to the High Uncertainty Rating.

We assign the firm a cost of equity of 7.6%. We do this because we see low risk to the overall business given its entrenched position, immunity to cyclicality, and strong financial position and margins. We see the business as low-risk and high-quality, while we see a high uncertainty related to our fair value based on future revenue growth compared with the overall semiconductor market.

Key risks for Cadence include headwinds in China, customer concentration, and operational execution risks related to future innovation.

Approximately 13% of total revenue comes from China. New regulations on exports to China, along with increasing competition in China's domestic semiconductor industry (China's long-term vision is for the Chinese market to consist of Chinese firms), could all pose headwinds to revenue growth.

Cadence depends on certain customers for large chunks of its revenue base, although it does business with largely all chip designers and has been diversifying into systems companies over time. Losing a major client could affect the company.

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Cadence faces execution risks related to future innovation, including the need to innovate to stay competitive with key rival Synopsys and to take share in new markets. The industry is always changing, and Cadence will have to keep up with these changes and new competitors.

We believe Cadence faces low levels of environmental, social, and governance risk.

Capital Allocation Eric Compton, CFA, Director, 28 Apr 2026

We assign Cadence an Exemplary Morningstar Capital Allocation Rating based on a sound balance sheet, exemplary investments regarding strategic priorities and execution, and appropriate shareholder distribution policies.

With solid profitability, low debt, and reasonable cash balances, we view the balance sheet as sound. With regards to shareholder distributions, Cadence prioritizes capabilities-focused M&A to further its overall competitive positioning and growth first, such as its recent acquisition of Hexagon's design and engineering segment, and then uses excess capital for reinvesting into the business and share repurchases. Share repurchases offset existing share issuance, and repurchases have been large enough to gradually reduce the share count over time. The company does not pay a dividend, although perhaps at some point in the future it might. We view all of this as reasonable.

Regarding investments, we view the company's moves as exemplary. Cadence has emerged from a fragmented EDA industry in the 1980s and 1990s and is now well positioned in an effective duopoly, due in no small part to all of the acquisitions and internal product innovation the company has done over the years. The company's recent acquisition of Hexagon's D&E segment further positions it for physical AI demand. We view these investments as exemplary because they have enabled Cadence to keep up with industry changes, build out a complete product portfolio, and add on additional business lines such as IP. All of these have increased shareholder value, and we think the company will continue to make such moves.

Cadence was formed from the 1988 merger of ECAD and SDA. Since then, it has gone through several CEOs and has engaged in numerous product launches and tuck-in acquisitions. We like that the company has developed key AI capabilities, including Cerebrus, which uses AI to help automate parts of the design process and also improve the performance of designs in ways that human engineers would have a hard time replicating on their own. Cadence is already launching AI agent tools, and we expect the firm will stay at the forefront of AI innovation, enabling its clients to get even more done. We also like that the company has built out its IP business over time, particularly its focus on the faster-growing, AI-linked IP blocks. Cadence has also built out its system design and analysis business in-house, including tools for computational fluid dynamics, electro-thermal, and other multi-physics effects. Solving these design issues will remain relevant and drive growth in the coming years.

Analyst Notes Archive

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Cadence Earnings: Strong Finish to 2025, While 2026 Outlook Shows Growth Will Remain Strong Eric Compton, CFA,Director,18 Feb 2026

Cadence reported revenue growth of 6% and adjusted EPS growth of 6% in the fourth quarter of 2025. Meanwhile, the 2026 outlook was solid, reaffirming the firm's ability to grow revenue by double digits annually with margin expansion. Why it matters: Cadence, along with seemingly every other software company, has sold off recently. We believe Cadence will be an AI beneficiary and is positioned uniquely well in an AI world. This quarter's results reaffirmed that belief. Management had a lot of good commentary on AI-related themes. Cadence is already deploying AI agents; it is seeing increased usage of its products, driven by AI, and AI is adding value for clients by improving productivity and results. Further, EDA firms like Cadence can charge for the added value and extra usage. We believe there is a lot of room for monetization of AI, with productivity going up as much as 10 times in certain workflows, and PPA improvements of 10% or more. The bottom line: We are maintaining our fair value of $330 per share for Cadence, or 40 times our adjusted 2026 EPS estimate, and view shares as moderately undervalued. We had thought shares were a bit pricey when they were trading above $350, but with the recent pullback, we are seeing some value. We made some tweaks to our 2026 estimates, but our final EPS estimate was essentially unchanged. We expect the 2026 guide to be likely conservative at the start of 2026, with room to rise as the year progresses, and our current estimates reflect that. Other highlights for us were the addition of 25 new logos to Cadence's digital design workflows, continued strength in IP, where the firm is taking share, along with a reaffirmation that the strong hardware sales cycle should continue. Long view: We expect that we are still early in the physical AI buildout, and we believe additional AI workflow monetization, along with new chip design efforts driven by physical AI demand, should power double-digit earnings growth for years to come.

Cadence Design Earnings: Positive Results Across the Board and China Contribution Back on Track Eric Compton, CFA,Director,28 Oct 2025

Cadence posted strong third-quarter results, with revenue up 10% year on year and non-GAAP operating margin coming in better than expected at 47.6%. China's revenue growth rate shot up to 53% year on year, and firmwide IP sales increased 10% over the previous year. Why it matters: Artificial intelligence megatrends and global infrastructure build-outs are fueling design activity and driving strong bookings and revenue for Cadence. Specifically, these secular tailwinds are driving the core electronic design automation business through higher demand for verification and hardware solutions. The company is seeing strong activity in China after the restrictions were lifted, as Chinese customers quickly caught up on their missed hardware purchases during the ban. The company also benefited from healthy underlying design activity and demand momentum from the region. We like the fact that Cadence's IP segment is posting solid growth, even while rival Synopsys struggles. Cadence's IP portfolio seems to be winning key memory-related mandates, displacing competitors, and doing well

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

with AI and high-performance computing workloads, all good signs. The bottom line: We raise our fair value estimate for wide-moat Cadence to $330 from $320 and are increasingly confident in Cadence to maintain its double-digit revenue growth. After adding together the strong results and updated full-year outlook, we view the shares as roughly fairly valued. The firm raised its guidance modestly on almost all key metrics, although the market seemed to have wanted even more, with the share price trending down slightly after the earnings release. We viewed shares as fully priced entering earnings, at roughly a 10% premium to our valuation. Management took a more optimistic tone while talking about China but also mentioned being prudent when incorporating the region's sales in the guidance. We still view slowing or declining sales from China as a risk worth watching.

Cadence Design Earnings: Beating Expectations as Demand Soars; China Business Back on Track
Eric Compton, CFA,Director,29 Jul 2025

Cadence Design delivered strong second-quarter results, with revenue and profitability coming above the guided range and bookings stronger than expected. Additionally, broad-based strength in other regions offset the impact of the near-month-long export restrictions on China. Why it matters: Cadence's revenue outperformance was driven by continued momentum in its electronic design automation tools, record-breaking hardware sales, solid IP usage, and standout double-digit growth in its system design and analysis segments. Even though China-based revenue took a short-term hit when US export controls paused shipments, trimming Chinese revenue from 11% to 9% sequentially, Cadence managed to recover losses by diversifying to other regions. Additionally, on a positive note, China orders have resumed. Non-GAAP operating margins exceeded the higher end of guidance, clocking in at 42.8%. We attribute this to a good operational execution and flow down of the robust demand the company is seeing for its solutions. The bottom line: We are raising our fair value estimate for wide-moat Cadence to $320 from $297, as we update our near-term growth expectations, given the strong results and a better-than-expected full-year outlook. As such, we continue to view the shares as slightly overvalued. We give credit for the strong bookings and the robust pipeline, which was the primary reason for the uplifted revenue guidance from a low-teens growth rate to now midteens growth. Furthermore, management now expects China-based revenue to increase, better than their earlier flat year-on-year outlook, and this is despite the month-long pause. This commentary was even more positive than we had expected. Coming up: For the upcoming quarter, we expect to see a robust renewal cycle and steady software demand that should help keep up the business momentum. We will not be surprised if backlog growth keeps trending upward, supported by demand for artificial intelligence tools like Cerebrus.

Synopsys and Cadence Design: EDA Firms Resume China Operations Following Trade Progress Eric Compton, CFA,Director,8 Jul 2025

Leading electronic design automation companies Synopsys and Cadence Design have received

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

approval from the US Department of Commerce to recommence their operations in China. Why it matters: China accounts for nearly 11% of revenue for each company. A month ago, the EDA companies were ordered to halt China operations indefinitely, which meant a substantial corporate revenue cut as business from China would have effectively been cut to $0. The news that both firms can now resume operations essentially brings them back on their original trajectory of flat year-on-year China growth for Cadence, and below corporate average growth for the China business for Synopsys. We believe this reversal comes on the back of the broader trade deal that the US and China are working toward. The bottom line: Based on this development, we update our fair value estimate for Synopsys to $550, up from $510; and $297 for Cadence, up from $275. We incorporates Cadence and Synopsys' original guidance for the coming quarter and fiscal year, as we no longer assume business from China drops to zero. We also take into account the loss of revenue for both the EDA firms during the China export restriction, specifically from the time both companies received letters to halt China operations (May 29) till the time they were able to restart business in China (July 2). We believe that the one-month interruption shaves a few basis points off this year's run-rate, but it does not alter our fundamental view on the companies or the EDA industry. Coming up: For Synopsys, the retraction of the China export restriction improves the prospects of Chinese regulatory approval for the merger with Ansys. Comparing our updated fair value estimate with the market price, Synopsys is slightly undervalued, as we believe the market has yet to fully price in a successful Ansys merger approval.

Ban on EDA Software Sales to China Could Be Coming Eric Compton, CFA,Director,29 May 2025

Reports have emerged suggesting that the Trump administration is asking semiconductor software design firms, known as electronic design automation, or EDA, to cease selling to China. Why it matters: Sales to China made up 11% of revenue in the most recent quarter for Cadence. We see an increased likelihood that those sales may be at risk. While there is some uncertainty about what has happened, we have seen multiple reports about notices received by EDA firms, including previous reporting on Siemens. Therefore, we think it likely that EDA firms have or will receive some notice, even if descriptions of the details vary. This is despite management from Synopsys denying having received an official notice on their recent earnings call. While we don't think management is being untruthful, we believe this is likely an indication of the overall direction in which the administration is headed, regardless of whether a notice is given today. The bottom line: We think the current series of events increases the probability above 50% that EDA sales to China will be banned, and it is now our base case. We are reducing our fair value estimate for Cadence to $275 per share from $300. We have also reduced our fair value estimate for Synopsys. Our fair value estimate assumes that sales to China drop to $0 by the end of the year and that China is roughly margin neutral. We acknowledge that there is a high degree of uncertainty and our fair value estimate could change again if there is no ban or if the margin profile deviates from our expectations. On the bright side, we view our updated valuations as "de-risked" from China exposure. Our updated valuation for Synopsys is $510, and we still view the shares

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

as moderately attractive. In contrast, we think shares of Cadence are slightly overvalued compared with our updated fair value estimate of $275.

Cadence Design Earnings: Good Results; Steady Growth Trajectory Amid Macroeconomic Uncertainty Eric Compton, CFA,Director,29 Apr 2025

Wide-moat Cadence Design Systems reported first-quarter results that were largely in line with management's guided range and our estimates, with no major surprises. After incorporating management's slightly higher full-year guidance into our model, we maintain our fair value estimate of $300 per share. At current prices, we see the company as fairly valued. We believe that electronic design automation firms tend to be more stable than the semiconductor industry in general, given their placement in the value chain, an attractive feature in the current turbulence.Chip design activity continues to be robust, and Cadence noted that it has not seen a change in customer behavior as semiconductor firms continue to invest in next-generation designs. Cadence's strong $6.4 billion backlog this quarter was up by roughly half a billion from the year before.One of the bigger questions is how the business in China will fare. While it is still early in the year, Chinese design activity remained robust in the quarter, with revenue in the region up 13%, an acceleration from the previous quarter. Cadence maintained its flat year-on-year growth outlook for the region, which we view as prudent given the current geopolitical uncertainty, although it does present upside in a more bullish scenario where everything turns out fine. Even with the good start to the year, we highlight that China-related revenue is higher-risk if the relationship between China and the US spirals out of control.We are seeing minimal impact from tariffs for Cadence, and management highlighted how it has strategically established multiple manufacturing lines within and outside the US, creating a diversified supply chain that minimizes tariff exposure on the hardware business, while the software and IP businesses are not subject to tariffs at all.

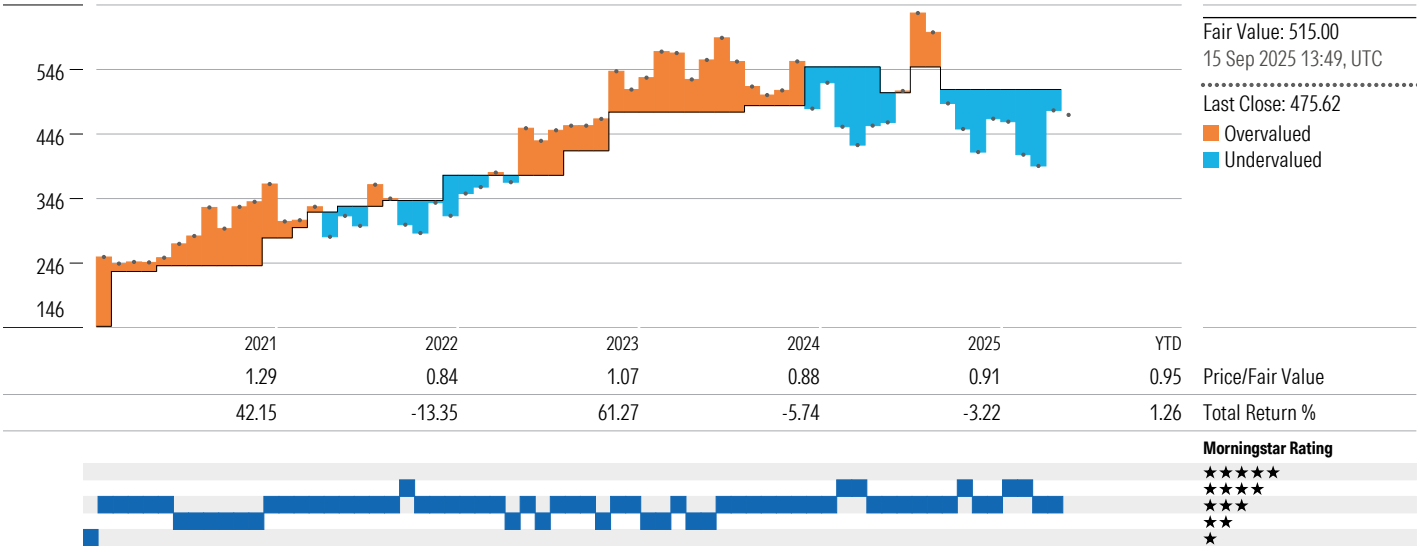
© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Competitors Price vs. Fair Value

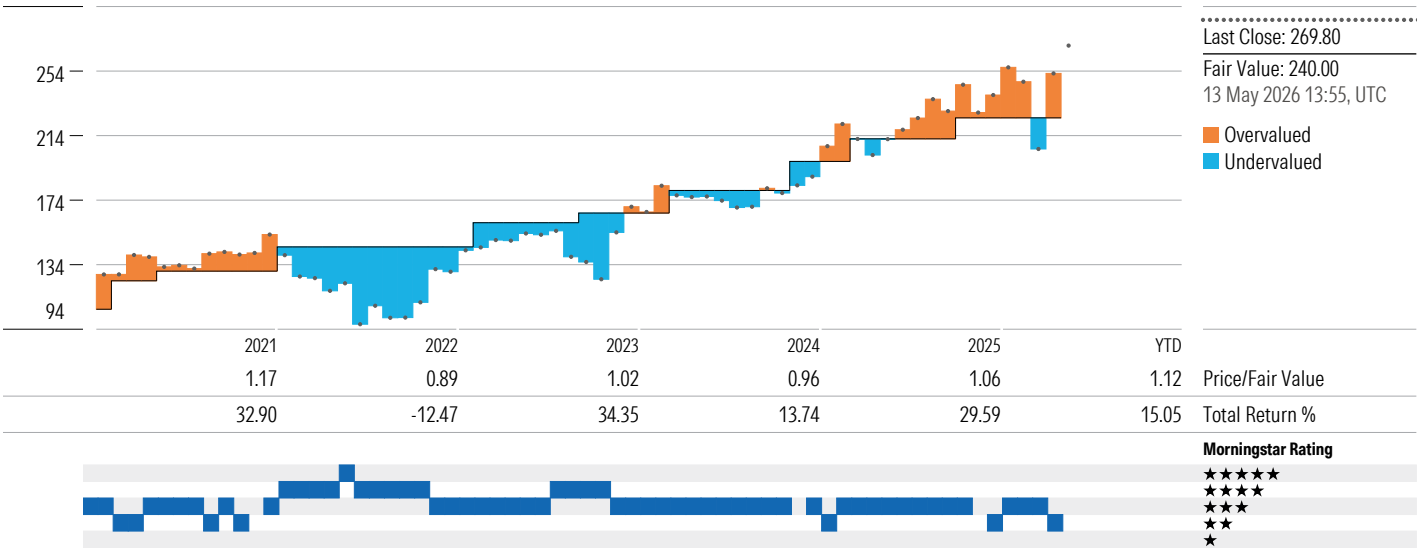
Synopsys Inc SNPS



	2021	2022	2023	2024	2025	YTD	
	1.29	0.84	1.07	0.88	0.91	0.95	Price/Fair Value
	42.15	-13.35	61.27	-5.74	-3.22	1.26	Total Return %

Total Return % as of 31 May 2026. Last Close as of 31 May 2026. Fair Value as of 15 Sep 2025 13:49, UTC.

Siemens AG SIE



	2021	2022	2023	2024	2025	YTD	
	1.17	0.89	1.02	0.96	1.06	1.12	Price/Fair Value
	32.90	-12.47	34.35	13.74	29.59	15.05	Total Return %

Total Return % as of 31 May 2026. Last Close as of 31 May 2026. Fair Value as of 13 May 2026 13:55, UTC.

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Morningstar Valuation Model Summary

Financials as of 27 Apr 2026

Fiscal Year, ends 31 Dec	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
Revenue (USD Mil)	4,090	4,641	5,297	6,214	7,030	7,889	8,817	9,813
Operating Income (USD Mil)	1,262	1,375	1,650	1,778	2,132	2,541	2,885	3,275
EBITDA (USD Mil)	1,397	1,548	1,720	2,201	2,635	3,112	3,156	3,572
Adjusted EBITDA (USD Mil)	1,817	2,110	2,535	3,153	3,768	4,217	3,974	4,419
Net Income (USD Mil)	1,041	1,055	1,109	1,245	1,524	1,884	2,153	2,440
Adjusted Net Income (USD Mil)	1,404	1,635	1,952	2,196	2,689	3,021	3,080	3,424
Free Cash Flow To The Firm (USD Mil)	600	-90	588	-600	1,763	2,036	1,994	2,287
Weighted Average Diluted Shares Outstanding (Mil)	273	274	273	276	278	278	279	279
Earnings Per Share (Diluted) (USD)	3.82	3.85	4.06	4.50	5.49	6.77	7.72	8.74
Adjusted Earnings Per Share (Diluted) (USD)	5.15	5.97	7.14	7.94	9.69	10.86	11.05	12.27
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Margins & Returns as of 27 Apr 2026

	3 Year Avg	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030	5 Year Avg
Operating Margin %	29.3	30.9	29.6	31.2	28.6	30.3	32.2	32.7	33.4	31.4
EBITDA Margin %	—	34.1	33.4	32.5	35.4	37.5	39.4	35.8	36.4	—
Adjusted EBITDA Margin %	—	44.4	45.5	47.9	50.7	53.6	53.4	45.1	45.0	49.6
Net Margin %	23.0	25.5	22.7	20.9	20.0	21.7	23.9	24.4	24.9	23.0
Adjusted Net Margin %	35.5	34.3	35.2	36.9	35.3	38.3	38.3	34.9	34.9	36.3
Free Cash Flow To The Firm Margin %	7.9	14.7	-1.9	11.1	-9.7	25.1	25.8	22.6	23.3	17.4

Growth & Ratios as of 27 Apr 2026

	3 Year CAGR	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030	5 Year CAGR
Revenue Growth %	14.1	14.8	13.5	14.1	17.3	13.1	12.2	11.8	11.3	13.1
Operating Income Growth %	15.4	17.6	8.9	20.0	7.7	19.9	19.2	13.5	13.5	14.7
EBITDA Growth %	12.6	15.8	10.8	11.1	28.0	19.7	18.1	1.4	13.2	16.1
Adjusted EBITDA Growth %	17.0	14.8	16.2	20.1	24.3	19.5	11.9	-5.8	11.2	11.8
Earnings Per Share Growth %	9.5	23.7	1.0	5.3	11.0	22.0	23.3	14.1	13.2	16.6
Adjusted Earnings Per Share Growth %	9.5	20.7	16.0	19.6	11.2	22.0	12.0	1.8	11.0	16.6

Valuation as of 27 Apr 2026

	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
Price/Earning	52.9	50.3	43.8	52.2	42.7	38.1	37.5	33.8
Price/Sales	18.1	17.7	16.0	18.4	16.2	14.5	13.0	11.6
Price/Book	21.8	17.6	15.6	15.6	12.4	10.1	8.4	7.1
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	40.6	39.0	33.4	36.8	30.8	27.5	29.2	26.2
EV/EBIT	58.5	59.9	51.3	65.2	54.4	45.6	40.2	35.4
Dividend Yield %	—	—	—	—	—	—	—	—
Dividend Payout %	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 27 Apr 2026

Fiscal Year, ends 31 Dec	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
ROA %	18.4	11.8	10.9	9.8	10.5	11.6	11.4	11.1
ROE %	30.6	22.6	20.3	16.9	16.5	16.5	15.7	14.9
ROIC %	24.3	20.8	20.5	18.2	17.6	19.3	20.3	20.9

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

Financial Leverage (Reporting Currency)	Actual			Forecast				
Fiscal Year, ends 31 Dec	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Debt/Capital %	1.0	3.0	3.0	2.4	2.0	1.4	1.3	1.4
Assets/Equity	1.7	1.9	1.9	1.7	1.6	1.4	1.4	1.3
Net Debt/EBITDA	-0.3	-0.1	-0.3	0.2	-0.6	-1.3	-1.9	-2.4
Total Debt/EBITDA	0.4	1.2	1.0	0.8	0.6	0.4	0.4	0.4
EBITDA/ Net Interest Expense	-59.2	-46.8	-84.5	33.8	104.7	203.1	643.1	353.1

Forecast Revisions as of 27 Apr 2026	2026		2027		2028	
Prior data as of 18 Feb 2026	**Current**	**Prior**	**Current**	**Prior**	**Current**	**Prior**
Fair Value Estimate Change (Trading Currency)	330.00	329.84	—	—	—	—
Revenue (USD Mil)	6,214	6,034	7,030	6,821	7,889	7,656
Operating Income (USD Mil)	1,778	1,946	2,132	2,265	2,541	2,626
EBITDA (USD Mil)	3,153	3,007	3,768	3,440	4,217	3,692
Net Income (USD Mil)	2,196	2,258	2,689	2,592	3,021	2,827
Earnings Per Share (Diluted) (USD)	4.50	5.09	5.49	5.91	6.77	7.10
Adjusted Earnings Per Share (Diluted) (USD)	7.94	8.26	9.69	9.45	10.86	10.29
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00

Key Valuation Drivers as of 27 Apr 2026

Cost of Equity %	7.6
Pre-Tax Cost of Debt %	5.5
Weighted Average Cost of Capital %	7.4
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	11.4
Stage II Investment Rate %	26.0
Perpetuity Year	20

Additional estimates and scenarios available for download at https://pitchbook.com/.

Discounted Cash Flow Valuation as of 27 Apr 2026

	USD Mil
Present Value Stage I	16,078
Present Value Stage II	21,367
Present Value Stage III	50,901
Total Firm Value	**88,346**
Cash and Equivalents	3,001
Debt	2,480
Other Adjustments	0
Equity Value	**88,868**
Projected Diluted Shares	275
Fair Value per Share (USD)	**330.00**

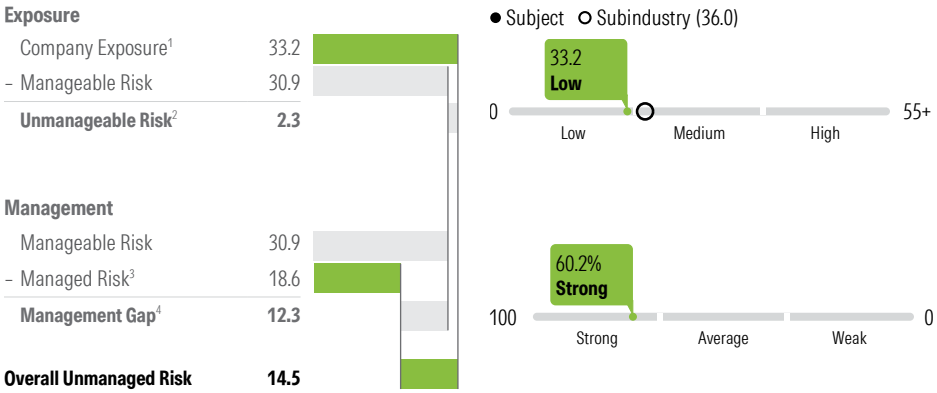
© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Cadence Design Systems Inc CDNS ★★ 1 Jun 2026 22:17, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
374.93 USD 31 May 2026	330.00 USD 28 Oct 2025 19:33, UTC	1.26	114.23 USD Bil 1 Jun 2026	Wide	Large Growth	High	Exemplary

ESG Risk Rating Breakdown



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

ESG Risk Rating is of May 30, 2026. Highest Controversy Level is as of May 08, 2026. Sustainalytics Subindustry: Enterprise and Infrastructure Software. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 60.2% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

Peer Analysis 30 May 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure			Management			ESG Risk Rating		
Cadence Design Systems Inc	33.2 | Low	0	55+	60.2 | Strong	100	0	14.5 | Low	0	40+
Siemens AG	58.2 | High	0	55+	66.4 | Strong	100	0	23.1 | Medium	0	40+
PDF Solutions Inc	39.2 | Medium	0	55+	36.3 | Average	100	0	25.8 | Medium	0	40+
Synopsys Inc	32.0 | Low	0	55+	61.3 | Strong	100	0	13.7 | Low	0	40+
AvidXchange Holdings Inc	31.2 | Low	0	55+	26.7 | Average	100	0	23.5 | Medium	0	40+

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Appendix

Historical Morningstar Rating

Cadence Design Systems Inc CDNS 1 Jun 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★	★★★	★★★	★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★	★★★	★★★	★★★	★★★	★★	★★	★★	★★	★★	★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★	★★	★★★	★★	★★	★★	★★	★★	★★	★★	★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★	★★★	★★★	★★★	★★	★★★	★★★	★★★	★★	★★	★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★	★	★★	★★	★★	★★	★★	★★	★★	★★	★★	★

Synopsys Inc SNPS 1 Jun 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★	★★★	★★★★	★★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★★	★★★	★★★	★★★	★★	★★	★★★	★★	★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★	★★★	★★★	★★★	★★	★★★	★★	★★★	★★★	★★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★★	★★	★★	★★	★★	★★★	★★★	★★★	★★★	★★★	★

Siemens AG SIE 29 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★	★★★	★★★	★★★	★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
—	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
—	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
—	★★	★★★	★★	★★★	★★★	★★★	★★★	★★	★★	★★★	★★★

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, indepth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss shortterm market-price movements), but we believe these negatives are mitigated by deep analysis and our longterm approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our singlepoint star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in workingcapital accounts, and capital spending. Based on these projections, we calculate earnings before interest, after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

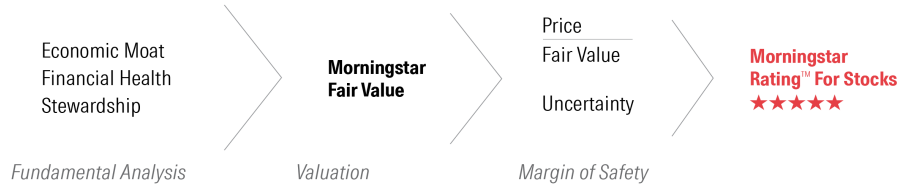
Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Research Methodology for Valuing Companies

outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

	Margin of Safety	
Qualitative Analysis Uncertainty Ratings	★★★★★Rating	★Rating
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

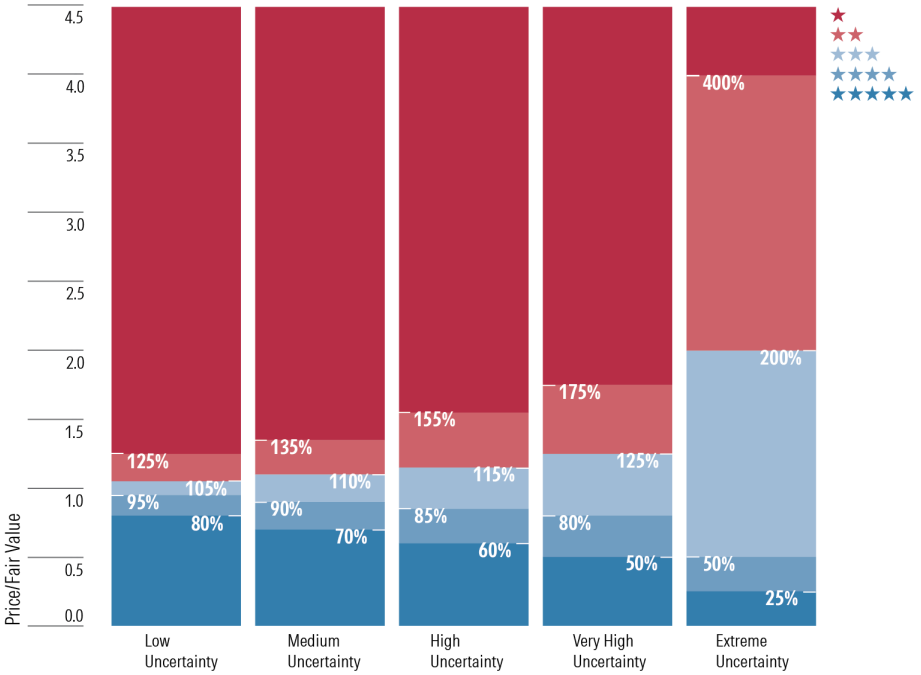
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to https://shareholders.morningstar.com

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multiyear time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Research Methodology for Valuing Companies

and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Sustainalytics ESG Risk Rating Assessment: The ESG Risk Rating Assessment is provided by Sustainalytics; a Morningstar company.

Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score. Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

Ratings should not be used as the sole basis in evaluating a company or security. Ratings involve unknown risks and uncertainties which may cause our expectations not to occur or to differ significantly from what was expected and should not be considered an offer or solicitation to buy or sell a security.

Risk Warning

Please note that investments in securities are subject to market and other risks and there is no assurance or guarantee that the intended investment objectives will be achieved. Past performance of a security may or may not be sustained in future and is no indication of future performance. A security investment return and an investor's principal value will fluctuate so that, when redeemed, an investor's shares may be worth more or less than their original cost. A security's current investment performance may be lower or higher than the investment performance noted within the report. Morningstar's Uncertainty Rating serves as a useful data point with respect to sensitivity analysis of the assumptions used in our determining a fair value price.

General Disclosure

Unless otherwise provided in a separate agreement, recipients accessing this report may only use it in the country in which the Morningstar distributor is based. Unless stated otherwise, the original distributor of the report is Morningstar Research Services LLC, a U.S.A. domiciled financial institution.

This Report is for informational purposes, should not be the sole piece of information used in making an investment decision, and has no regard to the specific investment objectives, financial situation or particular needs of any specific recipient. This publication is intended to provide information to assist investors in making their own investment decisions, not to provide investment advice to any specific investor. Therefore, investments discussed herein may not be suitable for all investors; investors must exercise their own independent judgment as to the suitability of such investments and recommendations in the light of their own investment objectives, experience, taxation status and financial position. Morningstar encourages Report recipients to read all relevant issue documents (e.g., prospectus) pertaining to the security concerned, including without limitation, information relevant to its investment objectives, risks, and costs before making an investment decision and when deemed necessary, to seek the advice of a financial, legal, tax, and/or accounting professional. The information, data, analyses and opinions presented herein are not warranted to be accurate, correct, complete or timely. Unless otherwise provided in a separate agreement, neither Morningstar, Inc. or the Equity Research Group represents that the report contents meet all of the presentation and/or disclosure standards applicable in the jurisdiction the recipient is located.

Except as otherwise required by law or provided for in a separate agreement, the analyst, Morningstar, Inc. and the Equity Research Group and their officers, directors and employees shall not be responsible or liable for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions within the report.

The Report and its contents are not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject Morningstar, Inc. or its affiliates to any registration or licensing requirements in such jurisdiction.

Where this report is made available in a language other than English and in the case of inconsistencies between the English and translated versions of the report, the English version will control and supersede any ambiguities associated with any part or section of a report that has been issued in a foreign language. Neither the analyst, Morningstar, Inc., or the Equity Research Group guarantees the accuracy of the translations.

This report may be distributed in certain localities, countries and/or jurisdictions ("Territories") by independent third parties or independent intermediaries and/or distributors ("Distributors"). Such Distributors are not acting as agents or representatives of the analyst, Morningstar, Inc. or the Equity Research Group. In Territories where a Distributor distributes our report, the Distributor is solely responsible for complying with all applicable regulations, laws, rules, circulars, codes and guidelines established

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

Research Methodology for Valuing Companies

by local and/or regional regulatory bodies, including laws in connection with the distribution third-party research reports.

Conflicts of Interest

- No interests are held by the analyst with respect to the security subject of this investment research report.
- Morningstar, Inc. may hold a long position in the security subject of this investment research report that exceeds 0.5% of the total issued share capital of the security. To determine if such is the case, please click http://msi.morningstar.com and http://mdi.morningstar.com
- Analysts' compensation is derived from Morningstar, Inc.'s overall earnings and consists of salary, bonus and in some cases restricted stock.
- Neither Morningstar, Inc. or the Equity Research Group receives commissions for providing research nor do they charge companies to be rated.
- Morningstar's overall earnings are generated in part by the activities of the Investment Management and Research groups, and other affiliates, who provide services to product issuers.
- Morningstar employees may not pursue business and employment opportunities outside Morningstar within the investment industry (including but not limited to, working as a financial planner, an investment professional or investment professional representative, a broker-dealer or broker-dealer agent, a financial writer, reporter, or analyst) without the approval of Morningstar's Legal and if applicable, Compliance teams.
- Neither Morningstar, Inc. or the Equity Research Group is a market maker or a liquidity provider of the security noted within this report.
- Neither Morningstar, Inc. or the Equity Research Group has been a lead manager or co-lead manager over the previous 12-months of any publicly disclosed offer of financial instruments of the issuer.
- Morningstar, Inc.'s investment management group does have arrangements with financial institutions to provide portfolio management/investment advice some of which an analyst may issue investment research reports on. However, analysts do not have authority over Morningstar's investment management group's business arrangements nor allow employees from the investment management group to participate or influence the analysis or opinion prepared by them.
- Morningstar, Inc. is a publicly traded company (Ticker Symbol: MORN) and thus a financial institution the security of which is the subject of this report may own more than 5% of Morningstar, Inc.'s total outstanding shares. Please access Morningstar, Inc.'s proxy statement, "Security Ownership of Certain Beneficial Owners and Management" section https://shareholders.morningstar.com/investor-relations/financials/sec-filings/default.aspx
- Morningstar, Inc. may provide the product issuer or its related entities with services or products for a fee and on an arms' length basis including software products and licenses, research and consulting services, data services, licenses to republish our ratings and research in their promotional material, event sponsorship and website advertising.

Further information on Morningstar, Inc.'s conflict of interest policies is available from http://global.morningstar.com/equitydisclosures. Also, please note analysts are subject to the CFA Institute's Code of Ethics and Standards of Professional Conduct.

Risk Warning Please note that investments in securities are subject to market and other risks and there is no assurance or guarantee that the intended investment objectives will be achieved. Past performance of a security may or may not be sustained in future and is no indication of future performance. A security's investment return and an investor's principal value will fluctuate so that, when redeemed, an investor's shares may be worth more or less than their original cost. A security's current investment performance may be lower or higher than the investment performance noted within the report. For investments in foreign markets there are further risks, generally based on exchange rate changes or changes in political and social conditions.

For more information about Morningstar's methodologies, please visit global.morningstar.com/equitydisclosures

For a list of securities which the Equity Research Group currently covers and provides written analysis on please contact your local Morningstar office. In addition, for historical analysis of securities covered, including their fair value estimate, please contact your local office.

For recipients in Australia: This Report has been issued and distributed in Australia by Morningstar Australasia Pty Ltd (ABN: 95 090 665 544; ASFL: 240892). Morningstar Australasia Pty Ltd is the provider of the general advice ('the Service') and takes responsibility for the production of this report. The Service is provided through the research of investment products.

To the extent the Report contains general advice it has been prepared without reference to an investor's objectives, financial situation or needs. Investors should consider the advice in light of these matters and, if applicable, the relevant Product Disclosure Statement before making any decision to invest. Refer to our Financial Services Guide (FSG) for more information at http://www.morningstar.com.au/fsg.pdf

For recipients in New Zealand: This report has been issued and distributed by Morningstar Australasia Pty Ltd and/or Morningstar Research Ltd (together 'Morningstar'). This report has been prepared and is intended for distribution in New Zealand to wholesale clients only and has not been prepared for use by New Zealand retail clients (as those terms are defined in the Financial Markets Conduct Act 2013).The information, views and any recommendations in this material are provided for general information purposes only, and solely relate to the companies and investment opportunities specified within. Our reports do not take into account any particular investor's financial situation, objectives or appetite for risk, meaning no representation may be implied as to the suitability of any financial product mentioned for any particular investor. We recommend seeking financial advice before making any investment decision.

For recipients in Hong Kong: The Report is distributed by Morningstar Investment Management Asia Limited, which is regulated by the Hong Kong Securities and Futures Commission to provide services to professional investors only. Neither Morningstar Investment Management Asia Limited, nor its representatives, are acting or will be deemed to be acting as an investment professional to any recipients of this information unless expressly agreed to by Morningstar Investment Management Asia Limited.

For recipients in India: This investment research is issued by Morningstar Investment Adviser India Private Limited. Morningstar Investment Adviser India Private Limited is registered with SEBI as a Portfolio Manager (registration number INP000006156) and as a Research Entity (registration number INH000008686). Morningstar Investment Adviser India Private Limited has not been the subject of any disciplinary action by SEBI or any other legal/regulatory body. Morningstar Investment Adviser India Private Limited is a wholly owned subsidiary of Morningstar Investment Management LLC. In India, Morningstar Investment Adviser India Private Limited has one associate, Morningstar India Private Limited, which provides data-related services, financial data analysis, and software development. The research analyst has not served as an officer, director, or employee of the fund company within the last 12 months, nor have they or their associates engaged in market-making activity for the fund company.The ESG-related information, methodologies, tool, ratings, data and opinions contained or reflected herein are not directed to or intended for use or distribution to India-based clients or users and their distribution to Indian resident individuals or entities is not permitted, and Morningstar/Sustainalytics accepts no responsibility or liability whatsoever for the actions of third parties in this respect.

*The Conflicts of Interest disclosure above also applies to relatives and associates of Manager Research Analysts in

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®

Research Methodology for Valuing Companies

India # The Conflicts of Interest disclosure above also applies to associates of Manager Research Analysts in India. The terms and conditions on which Morningstar Investment Adviser India Private Limited offers Investment Research to clients, varies from client to client, and are detailed in the respective client agreement.

For recipients in Japan: The Report is distributed by Ibbotson Associates Japan, Inc., which is regulated by Financial Services Agency, for informational purposes only. Neither Ibbotson Associates Japan, Inc., nor its representatives, are acting or will be deemed to be acting as an investment professional to any recipients of this information.

For recipients in Singapore: The Report is intended for Institutional Investor audiences and is distributed by Morningstar Investment Adviser Singapore Pte. Limited, which is licensed by the Monetary Authority of Singapore to provide financial advisory services in Singapore. Morningstar Investment Adviser Singapore Pte. Limited is the entity responsible for the creation and distribution of the research services described in this Report.

This content is provided for informational purposes only and may be shared or redistributed by Institutional Investors to their clients or other permitted persons, subject to obtaining the appropriate licence from Morningstar. Redistribution of this content is subject to any applicable conditions or limitations, including those agreed commercially or contractually with Morningstar. The person who shares or redistributes this content shall be solely responsible for compliance with all relevant legal and regulatory obligations in the jurisdictions in which the material is made available.

Investors should consult a financial adviser regarding the suitability of any investment product, taking into account their specific investment objectives, financial situation or particular needs, before making any investment decision.

Morningstar, Inc., and its affiliates rely on certain exemptions (Financial Advisers Regulations, Section 27(1)(e), Section 32B and 32C) to provide its investment research to recipients in Singapore.

For recipients in Korea: The report is distributed by Morningstar Korea Ltd., which has filed to the Financial Services Committee, for informational purposes only. Neither Morningstar Korea Ltd. nor its representatives are acting or will be deemed to be acting as an investment advisor to any recipients of this information.

© Morningstar 2026. All Rights Reserved. Unless otherwise provided in a separate agreement, you may use this report only in the country in which its original distributor is based. The information, data, analyses and opinions presented herein do not constitute investment advice; are provided solely for informational purposes and therefore are not an offer to buy or sell a security; and are not warranted to be correct, complete or accurate. The opinions expressed are as of the date written and are subject to change without notice. Except as otherwise required by law, Morningstar shall not be responsible for any trading decisions, damages or other losses resulting from, or related to, the information, data, analyses or opinions or their use. The information contained herein is the proprietary property of Morningstar and may not be reproduced, in whole or in part, or used in any manner, without the prior written consent of Morningstar. Investment research is produced and issued by subsidiaries of Morningstar, Inc. To order reprints, call +1 312-696-6100. To license the research, call +1 312-696-6869. Please see important disclosures at the end of this report.

MORNINGSTAR®